

CONTACT

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EDUCATION

2019-2026
YORK UNIVERSITY, CANADA
- PhD in Mathematics

2013-2015
NATIONAL CHENG KUNG UNIVERSITY, TAIWAN
- MSc in Applied Mathematics

2011-2013
UNIVERSITY OF UTAH, USA
- BSc in Mathematics

2009-2011
SNOW COLLEGE
- Associate of Science

LANGUAGES

- English (Fluent)
- Cantonese (Fluent)
- Mandarin (Fluent)
- Spanish (Basic)

(ALAN) HIN LON LAO

PhD, Mathematics

BACKGROUND

I am a citizen of China, Macao SAR. I was born and raised in Macao and began my professional mathematical journey in North America. My research focuses on matroid theory, especially transversal matroid theory. In particular, I work in identifying circuits of transversal matroids in their bipartite graph presentations, and finding a minimal presentation for a given presentation of a transversal matroid. Moreover, I am also interested in the upper approximation of non transversal matroids by transversal matroids.

My research is hiighly related to electric network theory and linear network coding theory for communication network.

I won the President's Award on poster competition in CMS (Canadian Mathematical Society) Meeting, Summer 2026.

RELATED WORK EXPERIENCE

York University, Canada 2019- 2026
Teaching assistant, research assistant
- Have been teaching assistant for Calculus, Linear Algebra, Discrete Mathematics for Computer Science, Introduction for Cryptography.
- Grading exams.
- Complete research goals under supervision.

Yueh Wah College(English Section), Macao 2018-2019
Mathematics teacher
- Teach mathematics in classroom setting, and provide individual advices to students if needed.
- Design curriculum with other math teachers.
- Corporate with teachers of other subjects to address student's needs.

Macao Sam Yuk School (English Section) 2017-2018
Mathematics resource teacher
- Provide one to one tutorial to students with special education needs.
- Provide in class education assistance if necessary.
- Corporate with homeroom and subject teachers .

Pui To Middle School 2016-2017
Mathematics teacher

National Cheng Kung University, Taiwan 2014-2015
Teaching assistant, research Assistant
- Took part in the research project: *Planar nearrings on the Euclidean Plane,* led by Prof. Ke, Wen Fong.
- Have been teaching assistant for Calculus, held tutorial session to address students questions once a week.
- Grading exams.

University of Utah 2012
Mathematics Tutor
- Worked in the math lab to answer questions of students having difficulty in mathematics.

CONFERENCES/SEMINARS TALKS

- *Acyclic Presentation of Transversal Matroid*. Student Research Session, Canadian Mathematical Society Meeting, University of New Brunswick, Saint John, NB. June 7th , 2026.
- *Transversal Matroids and their Presentations*. AARMS-CMS Student Poster Session, Canadian Mathematical Society Meeting, University of New Brunswick, Saint John, NB. June 6th , 2026.
- *Spotting Circuits in Bipartite Presentations of Transversal Matroids*. Contributed Talk, 29th Ontario Combinatorics Workshop, Queen's University, Kingston, ON. May 9th - 10th , 2026.
- *Directly Decomposing certain Relatively Complemented Lattices*. Contributed Talk, Ottawa Mathematics and Statistics Conference, 2024.
- *Random Matroid Polytope*. Student Talks, Mathematical Association of America Seaway Section, 2023 Spring.
- *The matroid represented by adjacency matrix of a graph*, The Algebraic Combinatorics Seminar, Fields Institute, 2022 Fall.

AWARD

- The President's Award of the Student Poster Competition, Canadian Mathematcial Society Summer Meeting 2026.
- York University PhD Fellowship, 2019-2026.
- Distinguished Overseas Chinese Student Scholarship, National Cheng Kung University, 2014-2015.

REFERENCE

Nantel Bergeron
Distinguished Research Professor
York Research Chair in Applied Algebra
Fellow of the Canadian Mathematical Society

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